MIL-S-51141A(MU)
12 September 1967
SUPERSEDING
MIL-S-51141(MU)
19 September 1963

MILITARY SPECIFICATION

SUIT, PROTECTIVE, IMPERMEABLE, SUPPLIED-AIR, M4

1. SCOPE

1.1 Scope. This specification covers one type and three sizes of a supplied air suit.

1.2 Classification. The suits shall be supplied in the following sizes as specified in the contract or order (see 6.2):

Small
Medium
Large

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-295 – Thread, Nylon.
UU-P-553 – Paper, Wrapping, Tissue.
ZZ-T-831 – Tubing, Rubber.
CCC-T-191 – Textile Test Methods.
PPP-T-60 – Tape, Pressure-Sensitive Adhesive, Waterproof for Packaging and Sealing.
PPP-B-601 – Boxes, Wood, Cleated-Plywood.
PPP-B-636 – Box, Fiberboard.
PPP-B-676 – Box, Setup.

Military

MIL-P-116 – Preservation, Methods of.
MIL-W-530 – Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Color.
MIL-C-12189 – Cloth, Coated, Butyl Coated, Toxicological Agents Protective.

STANDARDS

Federal

FED-STD-601 – Rubber: Sampling and Testing.
FED-STD-751 – Stitches, Seams, and Stitching.

Military

MIL-STD-105 – Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 – Marking for Shipment and Storage.

DRAWINGS

US Army Munitions Command

DL5-9-135 – Suit, Protective, Impermeable, Supplied-Air, M4

(All drawings referenced in this specification are listed on the above Edgewood Arsenal Drawing.)

PUBLICATIONS

Department of the Army Technical Manual

TM 3-4240-237-15 – Organizational, DS, GS, and Depot Maintenance Manual. Protective Outfit, Impermeable, Supplied-Air, M5.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Textile Color Card Association

United States Army Standard Color Card for the Official Standardized Shades of Sewing Thread.

(Copies can be obtained from the Textile Color Card Association, Inc., 200 Madison Ave., New York, N.Y. 10016.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 *Materials and components.*

3.1.1 *Materials.* All materials cited on Drawing DL5-9-135 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

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3.1.1.1 *Thread.* The thread shall conform to type I, class 1, size B, 3 ply of V-T-295. The thread shall be vat-dyed olive drab, shade S-1, cable 66022, of the U.S. Army Standard Color Card.

3.1.1.2 *Cement.* The cement used to seal the strapping over the seams shall be a self-curing neoprene cement, pigmented to match the color of the coated fabric. The cement shall not crack or flake when tested as specified in 4.2.3.1.

3.1.1.3 *Strapping.* The strapping used to seal the seams shall be bias-cut strips of cloth conforming to type I, class 1 of MIL-C-12189.

3.1.1.4 *Adhesion of strapping.* When tested as specified in 4.2.3.2(b), 4.2.3.2.(c), and 4.2.3.2.(d), the adhesion of strapping on all seams shall be at least 3.0, 2.0, and 2.0 pounds per inch of width, respectively.

3.1.1.5 *Butyl rubber compound.* The butyl rubber compound wrist and pant leg seals shall meet the physical requirements specified in table I.

Table I. Physical Requirements

Test	Requirement	Test paragraph
Tensile strength (pounds per square inch, minimum)		
Before aging	1600	4.3.4.2(a)
*After aging	20 percent maximum decrease from unaged	4.3.4.2(a)
Ultimate elongation (percent minimum)		
Before aging	450	4.3.4.2(b)
*After aging	20 percent maximum decrease from unaged	4.3.4.2(b)
Tensile stress at 200 percent elongation (pounds per square inch)	150-300	4.3.4.2(c)
Thickness (inches)	0.025 ± 0.005	4.3.4.2(d)

*Heat aging shall be in a circulating air oven at a temperature of 158° ± 2°F for a continuous period of 168 ± 1 hour or a temperature of 212° ± 2°F for a continuous period of 24 ± 1 hour as specified in FED-STD-601 (see 6.3).

3.1.2 *Components.* All components of the suit shall conform to the specifications and drawings listed on Drawing DL5-9-135 and subsidiary drawings.

3.1.2.1 *Wrist and leg seals.*

(a) *Manufacture.* The butyl rubber wrist and pant leg seals shall be manufactured by one of the following methods:

(1) Dip process
(2) Molding process
(3) Formed from calendered sheet

If method (3) is used, a longitudinal seam shall be made in the uncured sheet by rolling skived edges together before the bead end is rolled. The skived seam will minimize bulk when the cured seal is rolled up.

(b) *Wrist seals.* The wrist seals shall conform to the following details:

(1) Shape: Cylindrical, 4.000 ± 0.125 inch diameter

(2) Bead: One end rolled back to form bead 1/2 - 3/8 inch in diameter before cure

(3) Thickness: 0.025 ± 0.005 inch

(4) Length: 3.000 ± 0.125 inch

(c) *Pant leg seals.* The pant leg seals shall conform to the following details:

(1) Shape: Cylindrical with diameters as follows:

a. Small size: 5.125 ± 0.125 inch
b. Medium size: 5.375 ± 0.125 inch
c. Large size: 5.625 ± 0.125 inch

(2) Bead: One end rolled back to form bead 1/2 - 3/8 inch in diameter before cure.

(3) Thickness: 0.025 ± 0.005 inch

(4) Length: 3.000 ± 0.125 inch

3.2 *Manufacture and assembly.* The suit shall be manufactured and assembled as shown on figures 1 through 5 and on the subsidiary drawings which are listed on Drawing DL5-9-135.

3.2.1 *Stitches, seams, and stitching.* Stitches, seams, and stitching shall be in accordance with table II and shall conform to FED-STD-751. When stitching is not a continuous thread, each thread end or break shall be backstitched at least 2 inches whenever practicable. All seams and hems shall be backstitched at the ends to prevent raveling. The number of stitches per inch shall be no less than 8 nor more than 10. All joining seams shall be strapped. Seams and strapping shall be cleaned prior to application of cement, by wiping with naphtha or some similar cleaning agent.

3.2.2 *Marking.* The information listed below shall be stamped directly on the coated fabric of the suit below and to the left of the manifold assembly. The ink for marking shall be a vinyl base, black indelible type. Letters shall be sans-serif, extra bold, 14 point (caps only).

(a) Nomenclature (Suit, Protective, Impermeable, Supplied-Air, M4)

(b) Size of suit (small, medium or large)

(c) Number of this specification

(d) Name of contractor

3.2.3 *Assembly operations.* All operations listed in table II, III, IV and V shall be performed, but the sequence is not mandatory.

3.3 *Leakage.* When the suit is inflated with air to a pressure of 10 ± 1 millimeter of water, there shall be no evidence of leakage of air through any point on the surface of the suit for 30 seconds, except for seepage through the zipper closure, when tested as specified in 4.3.4.1.

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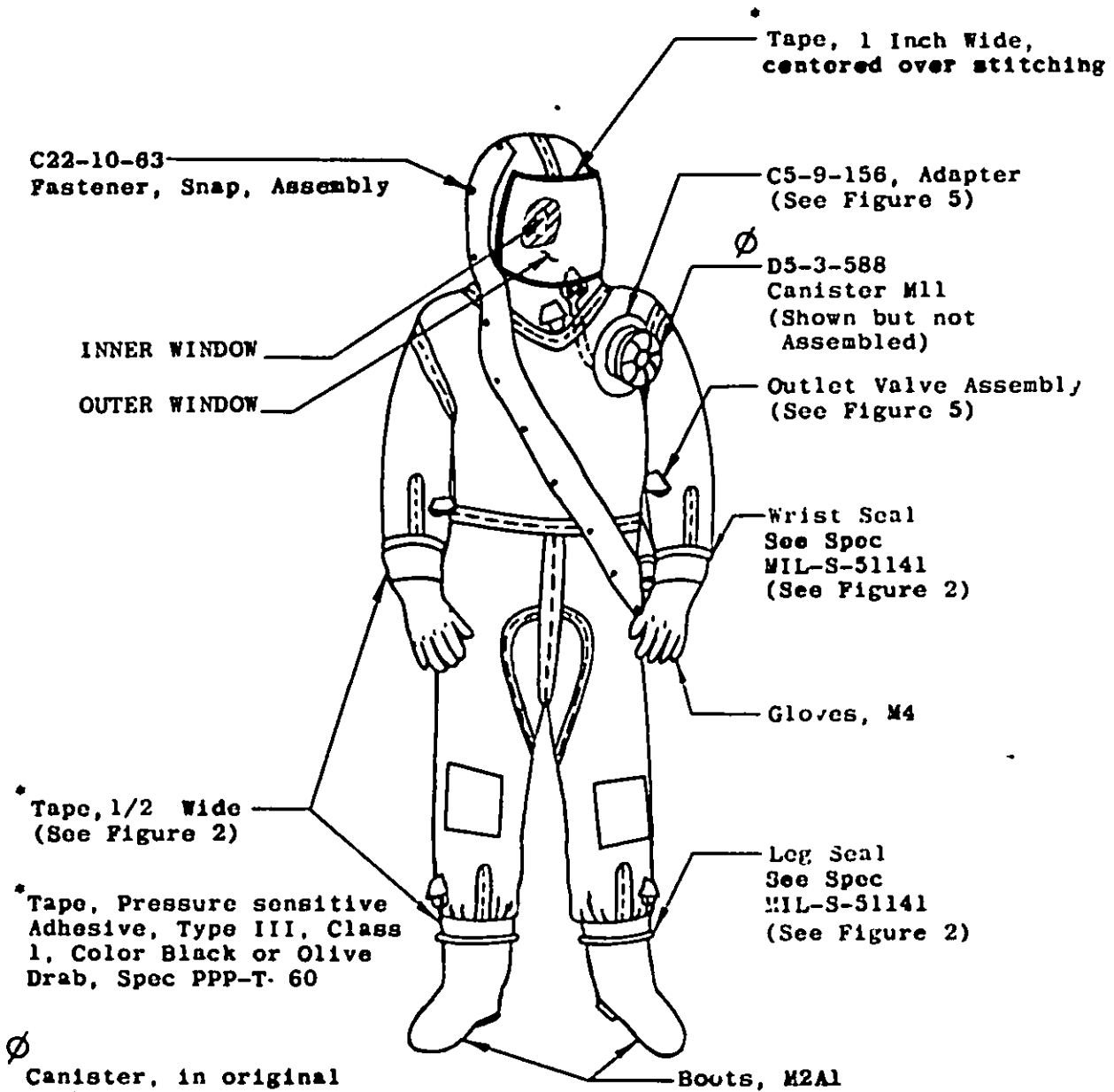


Figure 1 - Suit, Front

NOTE:
THE BOOTS AND GLOVES ARE SHOWN FOR ILLUSTRATIVE PURPOSES ONLY.

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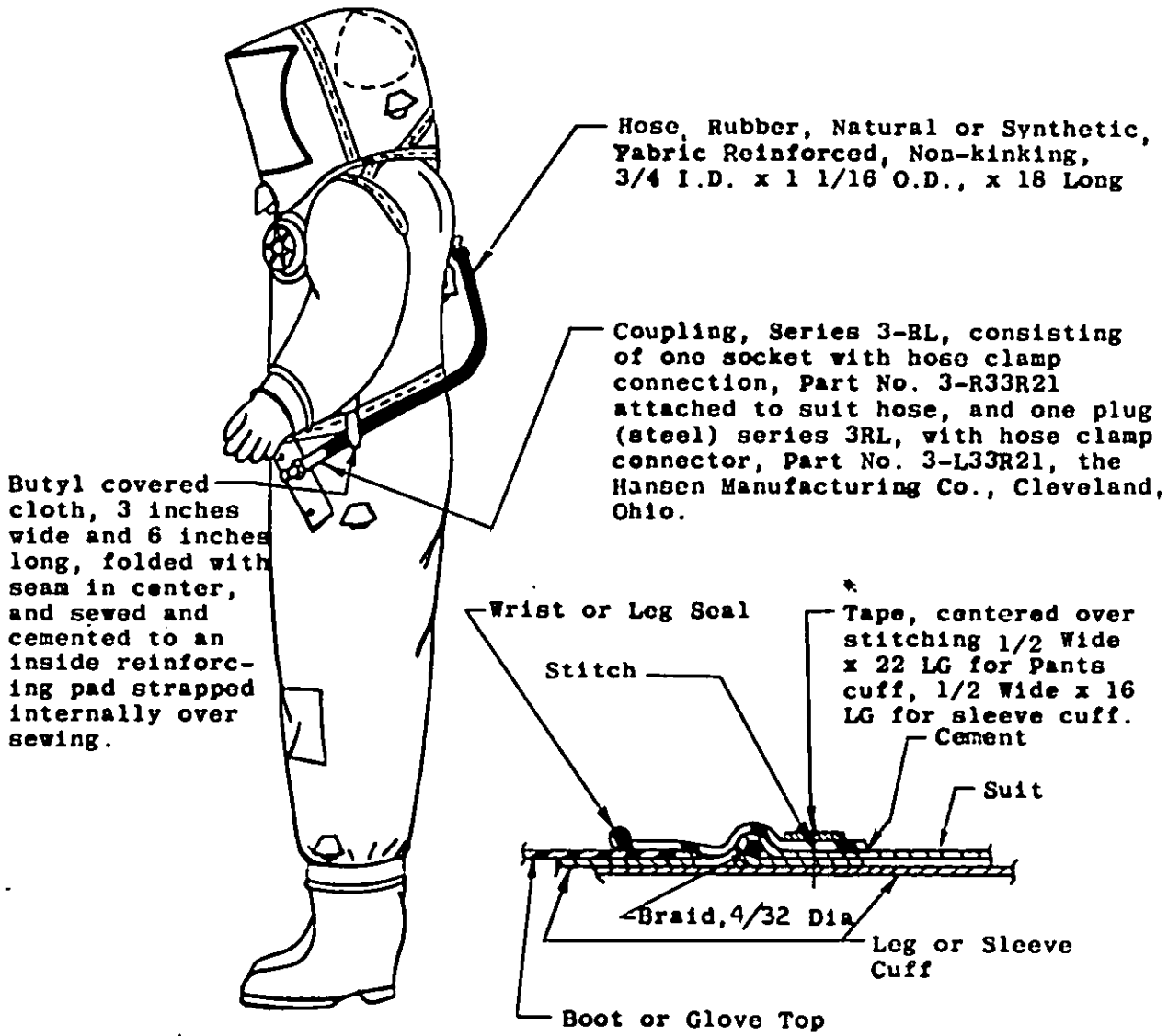


Figure 2 - Suit, Side

*Tape, pressure sensitive adhesive, type III, class 1, color black or olive drab, Spec PPP-T-60

NOTE:
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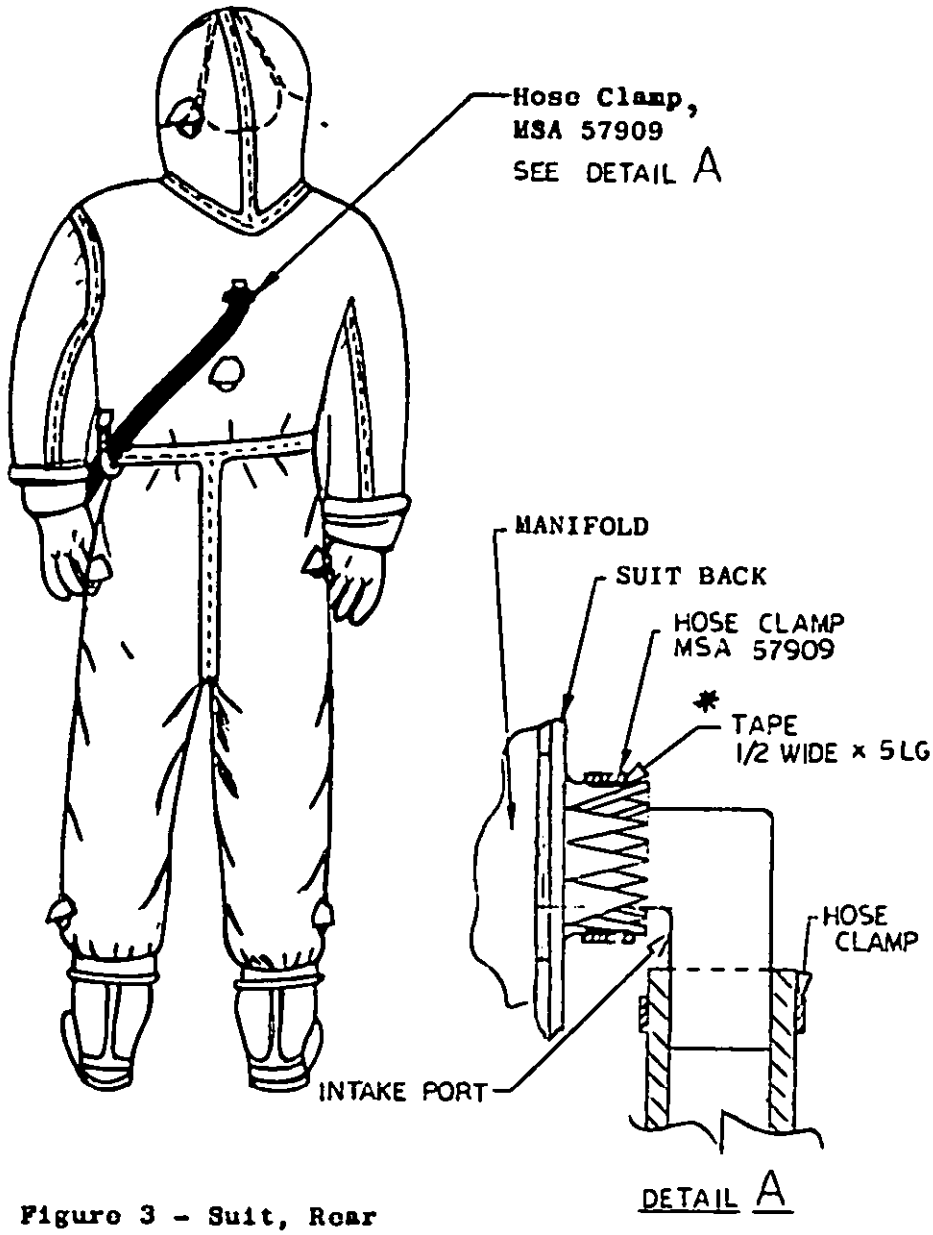


Figure 3 - Suit, Rear

* TAPE, PRESSURE SENSITIVE ADHESIVE, TYPE III, CLASS I, COLOR BLACK OR OLIVE DRAB, SPEC PPP-T- 60

NOTE:
THE BOOTS AND GLOVES ARE SHOWN FOR ILLUSTRATIVE PURPOSES ONLY.

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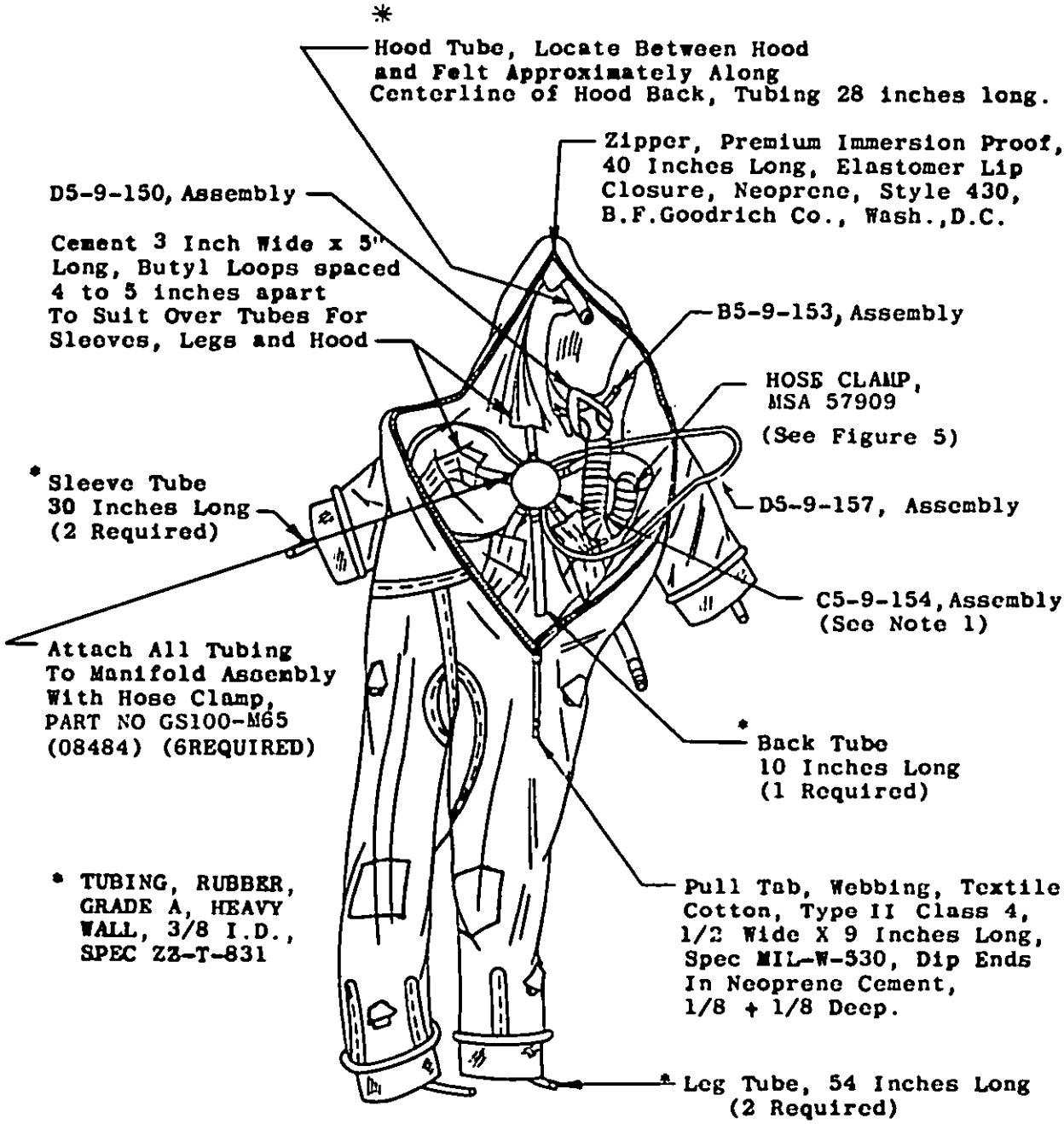
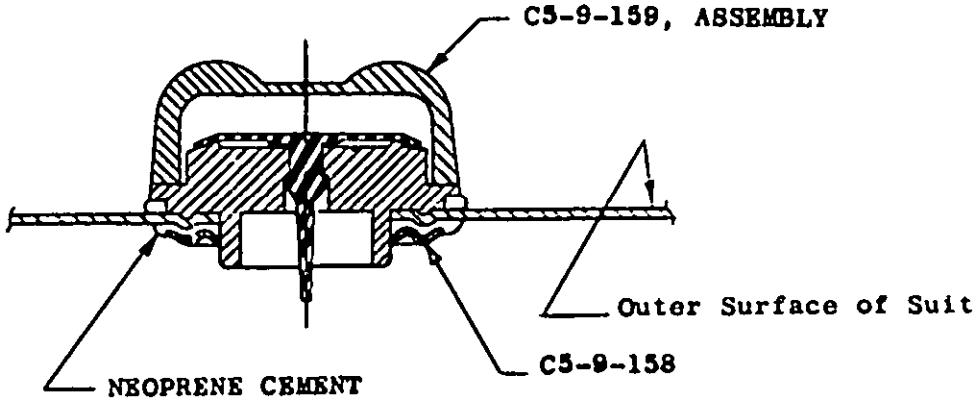


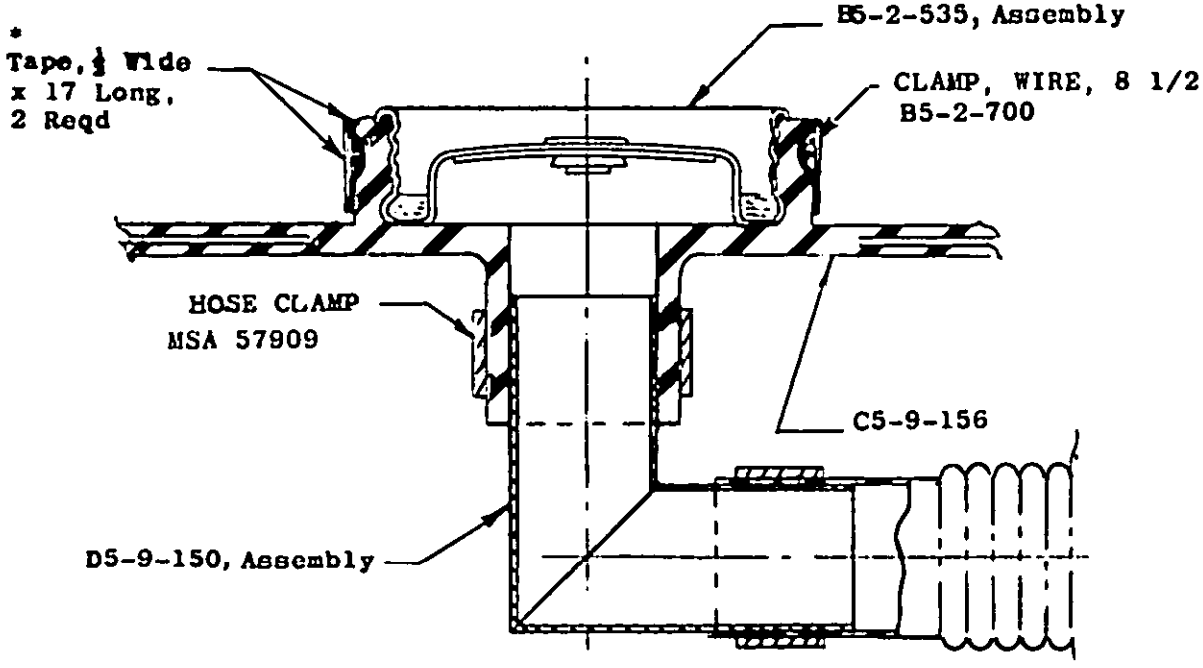
Figure 4 - Suit, Interior Arrangement

Note:
1. Locate C5-9-154 Assembly 1 3/4 Inches on CL of Back Piece Below Intersection of Hood Back. Cement Securely With Neoprene Cement.

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SECTION THRU OUTLET VALVE ASSEMBLY



SECTION THRU ADAPTER ASSEMBLY

Figure 5 - Sections through outlet valve and adapter

*Tape, pressure sensitive adhesive, Type III, class 1, color black or olive drab, Spec PPP-T-60

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Table II. *Assembly operations*

No.	Operation	Stitch type	Seam and stitching type
1	Stitch inner ply of window into inside of hood front	301	LSd-2
2	Join left back of hood to right back	301 or 401	LSb-2
3	Join front and back of hood	301 or 401	LSb-2
4	Strap inside and outside of all seams in hood	–	–
5	Tape outer ply of window in place, as shown in Figure 1, and tape wrapping paper cover to both sides of window to prevent scratching during remaining operations	–	–
6	Make pleats in back at shoulder and waist	301	SSa-1
7	Join front and back	301 or 401	LSb-2
8	Set sleeves with a double row of stitching, 3/16 to 5/16-inch wide, properly distributing fullness	301 or 401	LSb-2
9	Strap inside and outside of all seams in body	–	–
10	Join hood to body	301 or 401	LSb-2
11	Join sleeves and body seams	301 or 401	LSb-2
12	Join seat seam	301 or 401	LSb-2
13	Stitch crotch patch to front of pants	301	SSa-1
14	Join front of pants	301	LSb-2
15	Join body to pants	301	LSb-2
16	Stitch zipper in suit	301	LSd-2
17	Stitch flap over zipper	301	LSq-3
18	Join inseam of pants	301 or 401	LSb-2
19	Strap all seams	–	–
20	Stitch cord in extremities of sleeves and legs	301	EFa-2
21	Stitch plastic cuffs in sleeves and legs	301	LSa-2
22	Cement wrist seals and leg seals in place	–	–
23	Attach tubing to manifold outlets (see fig 4)	–	–
24	Cement manifold (Dwg C5-9-154) in top back of suit	–	–
25	Cement strips in legs, sleeves, and hood to form channels for tubing	–	–
26	Attach strap (Dwg D5-9-157) to manifold, and reeve the tubes through their channels	–	–
27	Cement and clamp the canister mounting (Dwg E5-2-535) in the adapter (Dwg C5-9-156), and tape the clamp (see fig 5)	–	–
28	Cut opening in upper left front of suit (see fig 1), and cement adapter in place	–	–
29	Attach outlet valve assemblies (see fig 5)	–	–
30	Stamp the label to the inside of the suit below and to the left of the manifold assembly	–	–
31	Cement felt padding into the hood portion	–	–

Table III. *Assembly of knee patch*

No.	Operation	Number required per inch
1	Cut 1/4" felt 7" x 9"	2 each
2	Cut butyl cloth 8" x 10-1/2"	2 each
3	Cut strap bias 1" x 9"	4 each
4	Cut strap bias 1" x 11-1/2"	4 each
5	Cement felt to knee location	2 each
6	Cover with butyl cloth by cementing edges to leg of suit	
7	Cement bias strap 1/2" over edges of patches to suit	

Table IV. *Assembly of outlet valve shield*

No.	Operation
1	Cut butyl cloth 5" x 5"
2	Apply cement to top of cloth cover
3	Position cover over valve at top forming 1/2" pleat on each side
4	Cement sides to suit
5	Cement bias straps 1/2" over edges of cover to suit

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Table V. *Harness*

No.	Operation	Number required
1	Cut 1/2" webbing 15" long Cut 1-1/2" webbing 5" long Cut 1-1/2" webbing 50" long	2 each 2 each 2 each
2	Sew slide to one end of 50" strap	
3	Thread loop to chafe, allow ends to meet	
4	Sew 3 rows of stitches to fasten loop to chafe	
5	Thread shoulder strap through loop on chafe and back through slide on shoulder strap	
6	Sew end on shoulder strap to 1/2" webbing strap, 1-3/4" from end, with longest point near end of strap	
7	Sew chafe 1-3/4" from shoulder strap with short end facing shoulder strap	
8	Sew end of loop on second chafe 1-3/4" from first chafe with long end facing first chafe	
9	Sew end of second shoulder strap 1-3/4" from second chafe loop with short end facing second chafe	
10	Attach fasteners, stud and eyelet 3/4" from end to center of fastener	
11	Attach fasteners, button, and socket 1-1/8" from end to center of fastener	

3.4 *Preproduction.* Prior to the start of regular production, a preproduction sample shall be produced in accordance with this specification for examination and test (see 4.2).

3.5 *Workmanship.* The suit shall be free from damage (*punctures, tears, or abrasions*) and foreign matter (dirt, grease, or oil).

4. QUALITY ASSURANCE PROVISIONS

4.1 *Responsibility for inspection.*

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 *Preproduction.*

4.2.1 *Sample.* A preproduction sample of two M4 protective suits of each size specified in the contract or order, six cement samples (see 4.2.3.1), nine strapped seam samples (see 4.2.3.2), and three each wrist and leg seals (see 3.1.2.1) shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production. Any change in type of cement, strapped seams, or butyl rubber compound for wrist and leg seals shall require a new preproduction sample.

4.2.2 *Inspection.*

4.2.2.1 *For examination and nondestructive test.* The preproduction sample M4 protective suits shall be examined in accordance with the classification of defects 4.3.3.4 and tested for leakage in accordance with 4.3.4.1.

4.2.2.2 *For destructive tests.* The cement samples shall be tested in accordance with 4.2.3.1, the strapped seam samples shall be tested in accordance with 4.2.3.2, and the butyl rubber compound wrist and leg seal samples shall be tested in accordance with 4.3.4.2.

4.2.3 *Tests.* Samples shall be prepared and tested as follows:

4.2.3.1 *Cement.*

(a) *Preparation of samples.* Using the technique to be followed in strapping the seams of the suit, apply two layers of cement to one side only of six samples of protective cloth conforming to type I, class 1 of MIL-C-12189. Each sample shall be 8-inches square and the cement coating shall completely cover one side of the sample. Dust the six samples tack-free after drying.

(b) *Before aging.* Allow three samples to remain at room temperature for a period of at least 72 hours. Condition the samples for 4 hours at -20° ±2°F and test in accordance with method 5874 of CCC-T-191, except that the hydrostatic test shall be omitted. In performing the test, consider the cemented side the "heavily coated side" of the specimen.

(c) *After aging.* Condition three samples at room temperature for at least 72 hours. Allow the samples to remain in a circulating-air oven for 168 consecutive hours at 158° ±2°F and then condition the samples for at least 4 hours at -20° ±2°F. Test in accordance with method 5874 of CCC-T-191, except that the hydrostatic test shall be omitted. In performing the test consider the cemented side the "heavily coated side" of the specimen.

4.2.3.2 *Strapped seams.*

(a) *Preparation of samples.* Fabricate nine 12-inch strapped seams using the same method, materials and equipment that will be used in regular production of the suits.

(b) *Before aging.* Cut a 6-inch specimen from the center of each of three samples and test in accordance with method 5962 of CCC-T-191 at least 48 hours after application of the strapping.

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